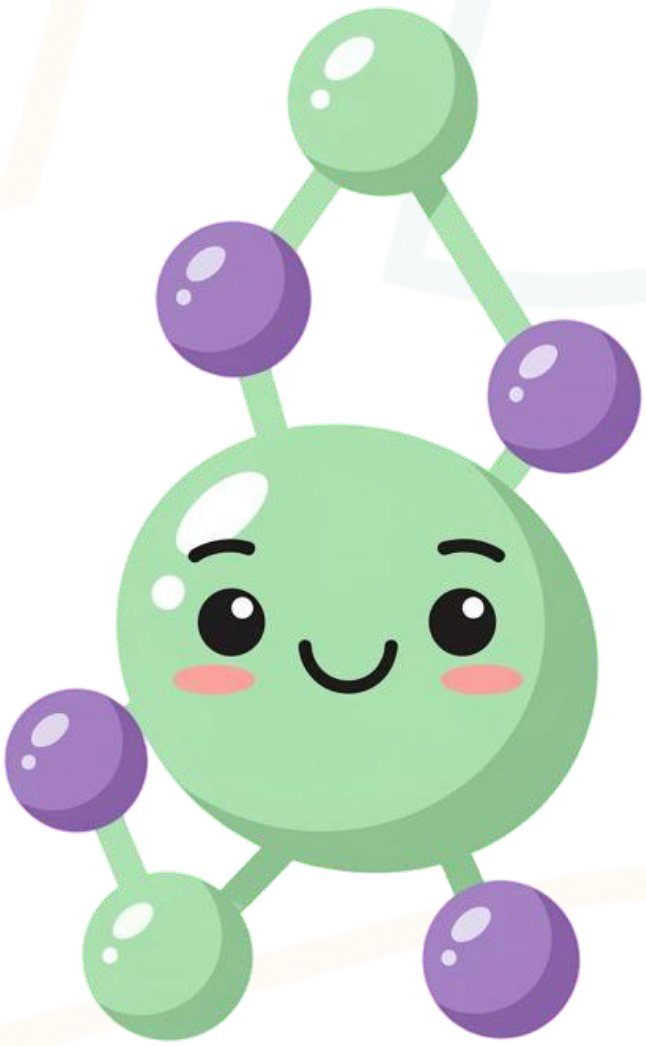
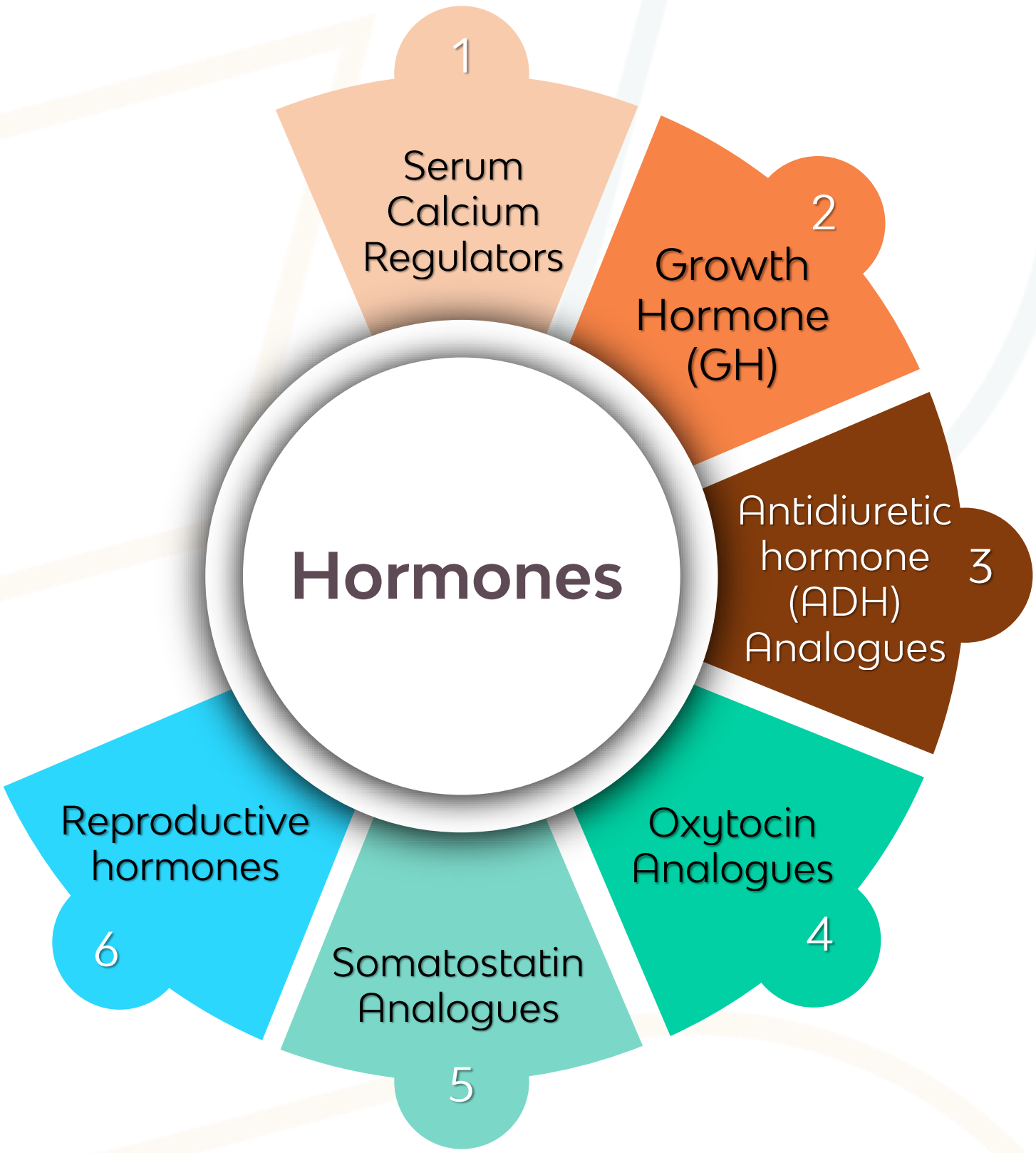
Hormones



Talent Management Department

صيدليـة المتحـدة
United Pharmacy

Hormones Drugs Classification



Today we will learn about

01 Serum Calcium Regulators

02 Growth Hormone (GH)

03 Antidiuretic Hormone (ADH) Analogues

04 Oxytocin Analogues

05 Somatostatin Analogues

06 Reproductive Hormones

Serum Calcium Regulators

Regulate calcium homeostasis in the blood and bones, involving three key hormones: Parathyroid hormone (PTH), vitamin D analogues, and calcitonin.

Mechanism of Action:
PTH analogues: Increase serum calcium by stimulating bone resorption and increasing renal reabsorption and intestinal absorption.

Vitamin D analogues: Enhance intestinal calcium absorption and support bone mineralization.

Calcitonin: Opposes PTH; inhibits osteoclast activity to reduce bone resorption. (Not currently available)

Uses: Osteoporosis, Hypocalcemia (e.g., in chronic kidney disease)
Vitamin D deficiency-related conditions

Note: Long-term use of PTH analogues is limited due to potential risk of osteosarcoma. Monitor calcium levels to avoid hypercalcemia.

1

Serum Calcium Regulators

Parathyroid Hormone Analogue

- **FORTEO®|** *Teriparatide*

Vitamin D 1α-hydroxy-cholecalciferol

- **ONE ALPHA®|** *Alfacalcidol*

Growth Hormone (GH)

Growth hormone regulates tissue growth and metabolism. Its analogues are used to manage disorders related to GH deficiency or excess.

Mechanism of Action:
GH analogues: Stimulate growth via IGF-1 production in the liver, promoting protein synthesis, bone growth, and lipolysis.

Uses:
GH deficiency in children and adults.

Contraindicated in active malignancy.

2

Growth Hormone (GH)

- NORDITROPIN®
- SAIZEN®

Antidiuretic hormone (ADH) Analogues

ADH analogues are synthetic versions of vasopressin used to manage fluid balance disorders.

Mechanism of Action:
Desmopressin (DDAVP): Selective V2 receptor agonist; promotes water reabsorption in renal collecting ducts without vasoconstriction.

Uses: Central diabetes insipidus, Nocturnal enuresis.

Must-know:
- Risk of water intoxication and hyponatremia.
- fluid intake must be monitored.

3

Antidiuretic hormone (ADH) Analogues

- **MINIRIN®|** Desmopressin "kidney Selective"

Oxytocin Analogues

Oxytocin analogues mimic the natural hormone oxytocin and are primarily used in obstetrics.

Mechanism of Action:
Stimulate uterine smooth muscle contraction via oxytocin receptors, and promote milk ejection from mammary glands.

Uses: Induction and augmentation of labor, Postpartum hemorrhage control.

4

Oxytocin Analogues

- **SYNTOCINON ®|** Oxytocin
  "Short Acting"
- **PABAL®|** Carbetocin
  "Long Acting"

Somatostatin Analogues

Somatostatin is a natural inhibitory hormone that suppresses secretion of various other hormones.

Mechanism of Action:
Bind to somatostatin receptors and inhibit the secretion of GH, insulin, glucagon, serotonin, and other GI hormones.

Uses: Acromegaly, Neuroendocrine tumors (e.g., carcinoid syndrome.

5

Somatostatin Analogues

- SANDOSTATIN ®| Octreotide

Reproductive Hormones

These hormones play a central role in controlling the hypothalamic-pituitary-gonadal (HPG) axis, affecting fertility, sexual development, and reproductive functions.

1. GnRH (Gonadotropin-Releasing Hormone) Agonists and Antagonists

GnRH is secreted in a pulsatile manner from the hypothalamus and stimulates the anterior pituitary to release FSH and LH

Agonists

Uses:

- Prostate and breast cancer (hormone-sensitive) – Endometriosis

Antagonists

Uses:

- Assisted reproductive technology (ART) to prevent premature LH surge.

⚠️ Monitor bone density with long-term use due to hypoestrogenism.

Reproductive Hormones

2. Gonadotropins – FSH & LH Analogues

FSH and LH are gonadotropins released from the anterior pituitary under the control of GnRH. They regulate ovarian follicular development, ovulation, and spermatogenesis

A. FSH Analogues

Stimulate growth and maturation of ovarian follicles in females, and spermatogenesis in males (via Sertoli cell activation).

Uses:

- Female infertility due to anovulation or polycystic ovary syndrome (PCOS)
- Male infertility (hypogonadotropic hypogonadism)
- In IVF protocols for ovarian stimulation.

B. LH Analogues – hCG (Human Chorionic Gonadotropin)

Stimulate final maturation of follicles and ovulation (in females), and testosterone production (in males, via Leydig cells).

Uses:

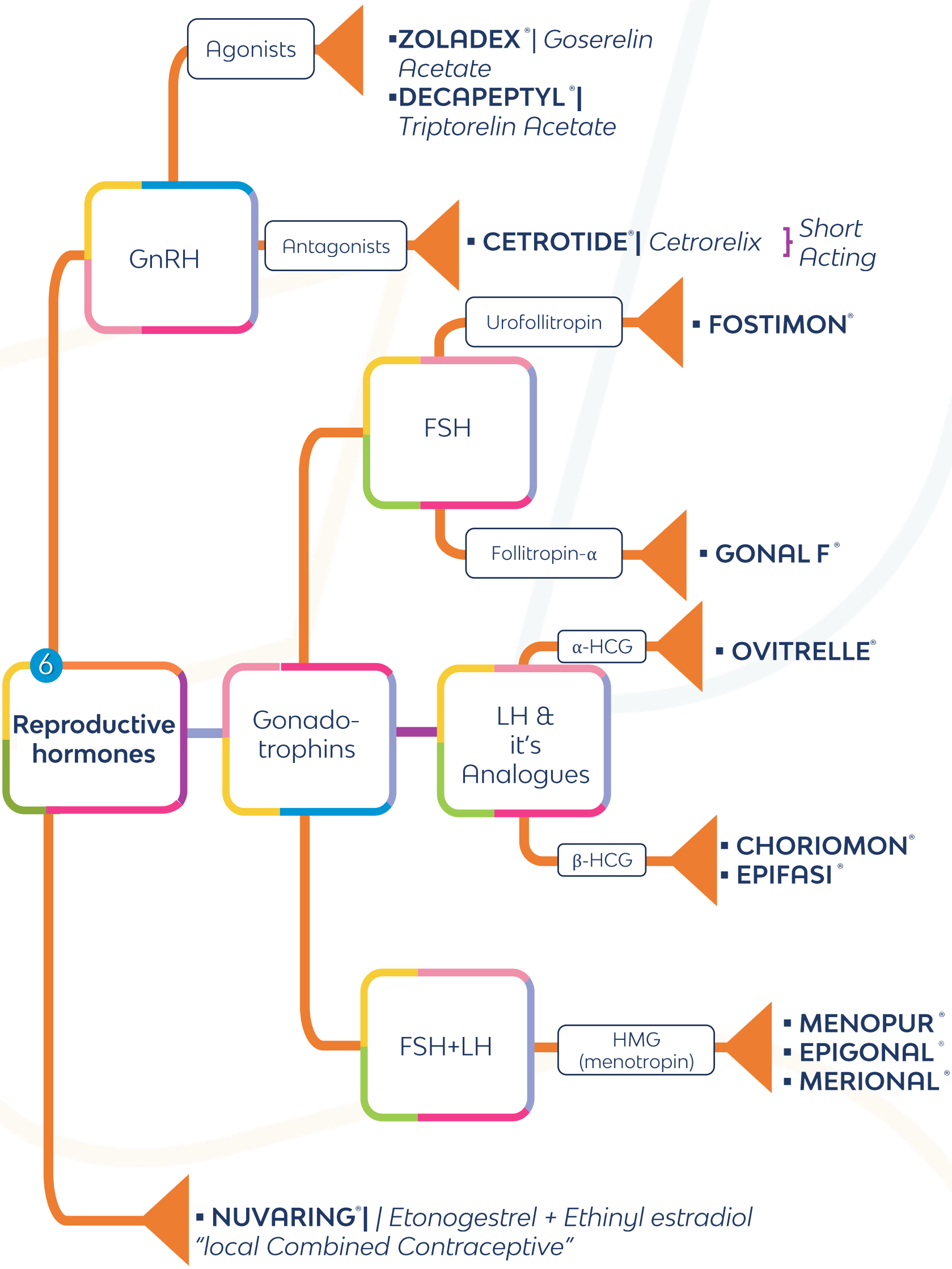
- Trigger ovulation after FSH stimulation
- Male infertility (stimulates testosterone)

Note: Beta-hCG subunit is used in pregnancy testing

C. Combined FSH + LH Preparations:

Uses:

- Female infertility in ART or ovulation induction
- Male infertility (stimulating spermatogenesis)


6
Reproductive hormones
GnRH
Agonists
▪ ZOLADEX® | Goserelin Acetate
▪ DECAPEPTYL® | Triptorelin Acetate
Antagonists
▪ CETROTIDE® | Cetrorelix
Short Acting
Gonado-trophins
FSH
Urofollitropin
▪ FOSTIMON®
Follitropin-α
▪ GONAL F®
LH & it's Analogues
α-HCG
▪ OVITRELLE®
β-HCG
▪ CHORIOMON®
▪ EPIFASI®
FSH+LH
HMG (menotropin)
▪ MENOPUR®
▪ EPIGONAL®
▪ MERIONAL®
▪ NUVARING® | Etonogestrel + Ethinyl estradiol
"local Combined Contraceptive"

الآن حان دور الأنشطة
اضغط على الزر للبدء



ابدأ النشاط